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Bode Integrals

The first integral applies to stable plants and the second to unstable plants. They are valid for every stabilizing controller, assuming only that both plant and controller have finite bandwidths. In words, the integrals state that the log of the magnitude of sensitivity function of a SISO feedback system, integrated over frequency, is constant. The constant is zero for stable plants, and it is positive for unstable ones. It becomes larger as the number of unstable poles increases and/or as the poles move farther into the right-half plane. (Technically, we must count all unstable poles here, including those in the compensator, if any.)

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = 0$$

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = \pi \sum_{p\in P} \mathrm{Re}(p)$$

Fact 1: Fundamental, Quantifiable Control Difficulty

One of the best-known examples of instability is the inverted pendulum, or the "broomstick balancing problem." Those of us who learned our craft in the 1950s and 1960s know this problem well, either through the little cart experiments that appeared during those years in various university labs around the world or through textbook examples and simulations. Recall that this problem was motivated by the space race. Control engineers had to learn how to balance rockets on top of their plumes on their way to earth orbit. It is still an important problem today, in the post-Challenger era, as modern boosters, now built in several countries around the world, become more difficult to balance.

I bring up the inverted pendulum because it nicely illustrates our first fact. The illustration is this: I can obviously balance an ordinary stable pendulum without difficulty. I can also easily balance a long inverted pendulum, even in front of a room full of people. However, I find it more difficult to balance a shorter inverted pendulum, and I find it impossible to balance a very short one. You have probably tried this yourself. The exact lengths you can balance might be different, but the trend will be the same.

Certainly the theorem just cited has an explanation for this illustration hidden in it somewhere, and we could try to extract that explanation using all the modern machinery at our disposal. Perhaps we could compute some minimum H_∞ norms or even some minimum structured singular values achievable by human controllers. Of course, in our calculations, we would need to pay due attention to the inherent handicaps of such controllers, such as reaction time, neuromuscular lags, limb inertias, and many other uncertainties covering the fact that humans, and most everything else in the physical world, are not finite dimensional, linear, and time invariant.

The Bode Integrals

I will try to explain these observations in the frequency domain the way Hendrik Bode might have done it. It turns out that such an explanation is insightful, plain, and clear and is therefore preferable to many modern ones.

First, note that the difference between balancing long sticks and short sticks has to do with the location of the unstable mode. This is obvious from the linearized equations of motion of the stick. Under the simplifying assumption that all mass is concentrated at the end of the stick, these equations show an unstable pole at $\sqrt{g/L}$, where g denotes the acceleration of gravity and L denotes length. Divergence becomes more rapid as L decreases.

A frequency domain quantification of control difficulty in the face of such changing instabilities is captured by the Bode integrals (see sidebar). At the risk of sounding dogmatic, I believe that every control theoretician and every control engineer should know these integrals and understand their meaning. Unfortunately, we have not always taught them well.

My own background illustrates this last observation. You can judge for yourself, using your own background. During my entire control education as an undergraduate and graduate student in the 1960s, I ran across only the first integral, only



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Figure 2. *Chernobyl nuclear power plant shortly after the accident on 26 April 1986.*

Bode 积分

第一个积分适用于稳定的被控对象，第二个积分适用于不稳定的被控对象。它们对每一个镇定控制器都成立，唯一的假设是被控对象和控制器的带宽均有限。用文字表述，这些积分说明，SISO 反馈系统灵敏度函数幅值的对数在频率上的积分为常数。对于稳定的被控对象，该常数为零；对于不稳定的被控对象，该常数为正。随着不稳定极点数量增加和/或极点向右半平面更深处移动，该常数会增大。（严格来说，这里必须计入所有不稳定极点，包括补偿器中可能存在的不稳定极点。）

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = 0$$

$$\int_0^\infty \ln|s(j\omega)|\,d\omega = \pi \sum_{p\in P} \mathrm{Re}(p)$$

事实 1：基本的、可量化的控制难度

最广为人知的不稳定性实例之一是倒立摆，或称“扫帚杆平衡问题”。我们这些在20世纪1950s和1960s学习控制技术的人对这个问题都很熟悉，无论是通过那些年世界各地大学实验室中出现的小车实验，还是通过教科书中的例子和仿真。回想一下，这个问题的研究动机源于太空竞赛。控制工程师必须学会如何在火箭飞往地球轨道的过程中，使其在喷流上方保持平衡。如今，在“挑战者”号事故之后的时代，这仍然是一个重要问题，因为目前由世界多个国家制造的现代助推器越来越难以保持平衡。

我提到倒立摆，是因为它很好地说明了我们的第一个事实。具体来说：显然，我可以毫不费力地使一个普通的稳定摆保持平衡。我也可以轻松地使一个长倒立摆保持平衡，即使面对满屋子的人也不例外。然而，我发现使较短的倒立摆保持平衡更加困难，而使非常短的倒立摆保持平衡则是不可能的。你很可能也亲自尝试过。你能使之保持平衡的具体长度或许有所不同，但趋势是一样的。

当然，刚才引用的定理中必定隐含着对这一现象的解释，我们可以尝试运用我们掌握的所有现代方法来提取这一解释。也许我们可以计算人类控制器所能实现的某些最小 H_∞ 范数，甚至某些最小结构奇异值。当然，在计算中，我们需要充分考虑这类控制器固有的局限，例如反应时间、神经肌肉滞后、肢体惯性，以及许多其他不确定性；这些不确定性涵盖了这样一个事实：人类以及物理世界中的绝大多数其他事物都不是有限维、线性且时不变的。

波德积分

我将尝试以 Hendrik Bode 可能采用的方式，在频域中解释这些观察结果。事实证明，这样的解释富有洞见、浅显而清晰，因此比许多现代解释更可取。

首先，请注意，平衡长棍与短棍的差别与不稳定模态的位置有关。这一点从棍的线性化运动方程中显而易见。在所有质量都集中于棍端的简化假设下，这些方程表明，不稳定极点位于 $\sqrt{g/L}$，其中 g 表示重力加速度，L 表示长度。发散会随着 L 的减小而加快。

面对这种不断变化的不稳定性，波德积分给出了控制难度的频域量化描述（见边栏）。尽管这样说可能显得武断，我仍认为每一位控制理论研究者和控制工程师都应当掌握这些积分，并理解其含义。遗憾的是，我们并非总能把它们讲授清楚。

我自己的背景说明了最后这一观察。你可以根据自己的背景自行判断。在 1960s，我作为本科生和研究生接受控制学教育的整个过程中，只接触过第一个积分，而且仅仅



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图 2。*1986* 年 *4* 月 *26* 日事故发生后不久的切尔诺贝利核电站。

and distributing mass appropriately. In the years before full-authority automatic flight controls, most airplane designs constrained the placement of these elements to ensure stability over all flight regimes and all loading conditions. (The Wright brothers' airplane is a notable exception.) Today, however, when automatic controls are accepted and trusted in so many applications, there are important benefits to be gained by making the basic design unstable. For instance, note that a stable airplane requires a tail to balance the moment produced by lift acting through the cp – cg offset. The force produced by the tail is down, opposing the lift and making the overall configuration aerodynamically less efficient. The airplane also needs to carry the weight of the tail, which reduces payload. The less stable an airplane is, the smaller these performance and weight penalties. There are also other reasons for wanting instability, having to do with maneuverability and speeds of command response.

In the case of the X-29, the benefits of instability were desired in the transonic and supersonic flight regimes, so the airplane was designed to be modestly unstable in those regimes. Unfortunately, there is a basic aerodynamic phenomenon that moves the center of pressure of a lifting surface dramatically aft as speeds go from sub- to supersonic. As a result, the X-29's slight instability at supersonic speeds turns into a much more dramatic instability at subsonic speeds. Indeed, the airplane's real pole is as large as +6 rad/s in some flight regimes.

Recalling the broomstick's formula ($p = \sqrt{g/L}$), flying this airplane manually corresponds to balancing a 1-ft-long stick. It is not something we can expect a pilot to do without automatic assistance, at least for very long.

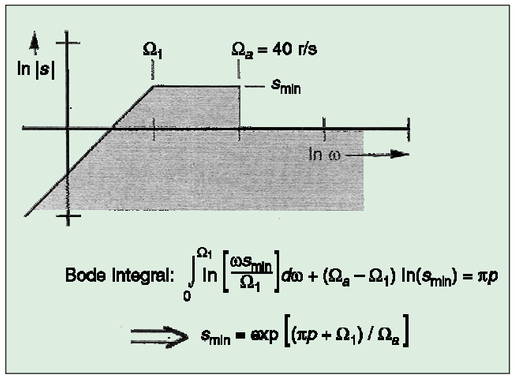


Figure 7. *Prototype X-29 sensitivity function.*

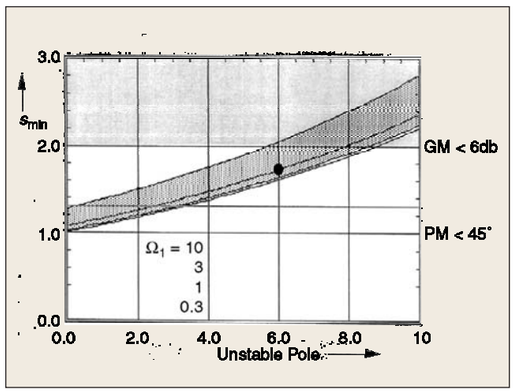


Figure 8. *Minimum X-29 sensitivities.*

X-29 Available Bandwidth

As with the broomstick, control difficulties associated with the X-29's unstable pole should not arise merely because the pole is large. Rather, they should arise if the pole is large compared with available bandwidth. Some of the major hardware elements in the control loop that limit this bandwidth include the following:

- *Sensors*: rate gyros and accelerometers, used for inner-loop stabilization. Their bandwidths, measured in the usual 3-dB-gain sense, are typically 120 rad/s or more.
- *Control processors*: digital computer systems sampling sensor data and computing surface commands at 80 Hz. This update rate can pass signals with good fidelity up to 30-40 rad/s (two to three samples per radian).
- *Actuators*: high-pressure hydraulic systems with servos to position each aerodynamic control surface. For the pitch axis, control surfaces include the canard, inboard and outboard flaperons along the wing, and strakes at the tail. The servo bandwidths, again measured by 3-dB gain, are approximately 20 rad/s. Basically, each of the servos is a position feedback loop around a hydraulic ram (an integrator). This gives a dominant first-order-lag response that retains its characteristics up to perhaps 70-80 rad/s, where valve dynamics, fluid compressibility, and local structural dynamics begin to take effect. The bottom line is that the available bandwidth of actuators is approximately 70 rad/s, at least for small signals.
- *Aerodynamics*: flow conditions around the airplane that map its geometric configuration (e.g., net orientation and control surface positions) into forces and moments. At low frequencies, this map is algebraic and well known, but because air itself has mass and momentum, the map becomes dynamic and poorly known on time scales comparable to characteristic length divided by velocity. For a 2-ft surface traveling at 200 ft/s, that time scale is 0.01 s, so the available bandwidth for aerodynamics is approximately 100 rad/s.
- *Airframe:* the mechanical structure linking the attachment points of actuators to the attachment points of sensors. (These are the important structural points affecting control. Other disciplines are, of course, interested in additional attachment points, such as where the wings are attached, where the pilot sits, etc.) The structure is rigid at low frequencies, but it begins to bend and flex dramatically at higher fre-

并合理分配质量。在全权限自动飞行控制出现之前，大多数飞机设计都对这些要素的布置加以限制，以确保在所有飞行状态和所有载荷条件下保持稳定。（莱特兄弟的飞机是一个显著的例外。）然而，如今自动控制已在众多应用中得到接受和信任，使基本设计具有不稳定性能够带来重要益处。例如，请注意，稳定的飞机需要尾翼来平衡升力通过 cp–cg 偏移产生的力矩。尾翼产生的力向下，与升力方向相反，使整体构型的气动效率降低。飞机还需要承载尾翼的重量，从而减少有效载荷。飞机的稳定性越低，这些性能损失和重量代价就越小。希望飞机具有不稳定性还有其他原因，涉及机动性和指令响应速度。

对于 X-29，希望在跨声速和超声速飞行状态下获得不稳定性带来的益处，因此该飞机被设计为在这些飞行状态下具有适度的不稳定性。遗憾的是，存在一种基本的空气动力学现象，它会使升力面的压力中心

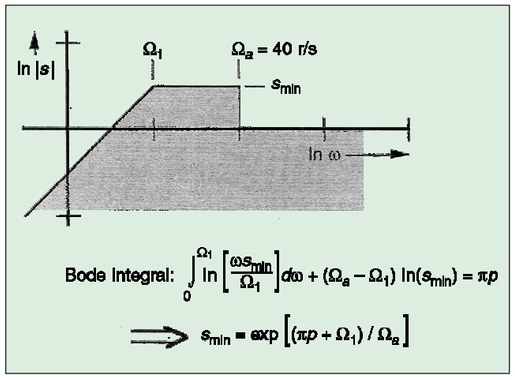


图 7。*X-29* 原型机的灵敏度函数。

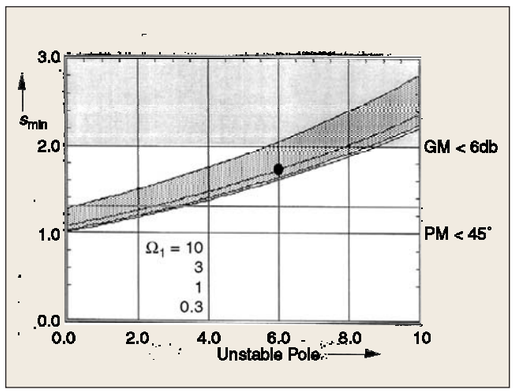


图 8。*X-29* 的最小灵敏度。

随着速度从亚声速增至超声速而大幅后移。因此，X-29 在超声速下的轻微不稳定性，在亚声速下会变为严重得多的不稳定性。事实上，在某些飞行状态下，该飞机的实极点高达 +6 rad/s。

回顾扫帚杆的公式(/)*p g L*= ., 手动驾驶这架飞机相当于使一根 1-ft 长的杆保持平衡。我们不能指望飞行员在没有自动辅助的情况下做到这一点，至少不能指望其长时间做到。

X-29 可用带宽

与扫帚杆一样，与 X-29 的不稳定极点相关的控制困难不应仅仅因为该极点很大而产生。相反，当该极点相对于可用带宽较大时，才会产生这些困难。控制回路中限制该带宽的一些主要硬件元件包括：

• 传感器：速率陀螺仪和加速度计，用于内环稳定。按通常的 3-dB 增益标准衡量，其带宽通常为 120 rad/s 或更高。• 控制处理器：以 80 Hz 采样传感器数据并计算舵面指令的数字计算机系统。这一更新速率能够以良好的保真度传递高达 30-40 rad/s 的信号（每弧度两到三个采样点）。

• 执行器：配有伺服机构的高压液压系统，用于定位各个气动操纵面。对于俯仰轴，操纵面包括鸭翼、沿机翼分布的内侧和外侧襟副翼，以及尾部边条。伺服机构的带宽同样以 3-dB 增益衡量，约为 20 rad/s。基本上，每个伺服机构都是围绕液压作动筒（积分器）构成的位置反馈回路。这使其响应以一阶滞后为主，并可保持这一特性直至大约 70-80 rad/s，此时阀门动态、流体可压缩性以及局部结构动态开始起作用。归根结底，执行器的可用带宽约为 70 rad/s，至少对小信号而言如此。

• 空气动力学：飞机周围的流动条件，将其几何构型（例如整体姿态和操纵面位置）映射为力和力矩。在低频下，这种映射是代数关系，且已得到充分认识，但由于空气本身具有质量和动量，在与特征长度除以速度相当的时间尺度上，这种映射会呈现动态特性，且人们对其认识不足。对于以 200 ft/s 运动的 2-ft 翼面，该时间尺度为 0.01 s，因此空气动力学的可用带宽约为 100 rad/s。

• 机体：连接执行器安装点与传感器安装点的机械结构。（这些是影响控制的重要结构点。当然，其他学科还关注其他安装点，例如机翼的连接位置、飞行员的座位位置等。）该结构在低频下呈刚性，但在较高频率下开始发生显著的弯曲和挠曲